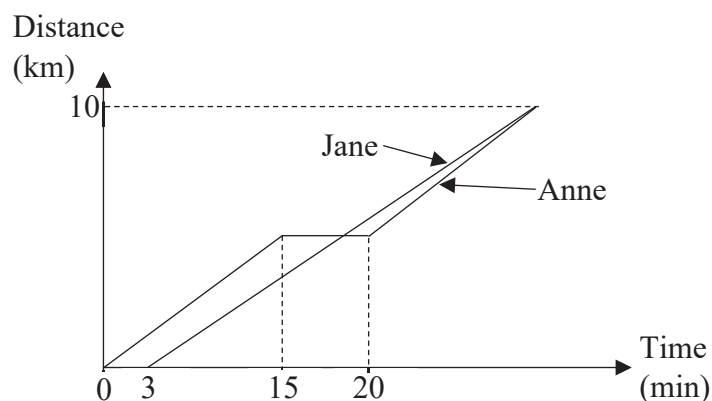


- 18 The figure below shows the distance-time graph of Anne's 10 km roller skating journey along East Coast Park.



Anne skates at an average speed of 0.3 km/min for the first 15 min. She takes a 5-min break and continues with an average speed of 0.5 km/min to the end point.

- (a) Find the distance travelled by Anne in the first 15 min.

Answer km [1]

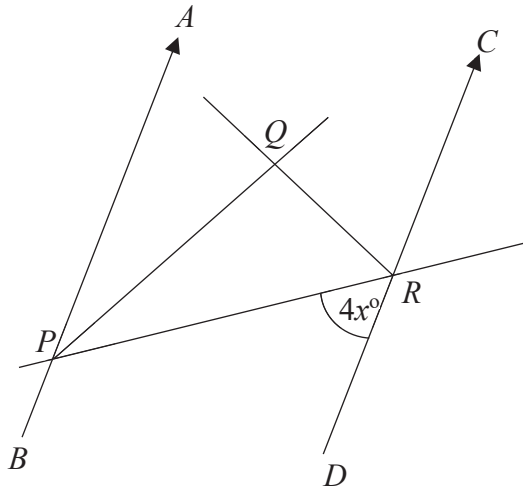
- (b) Find the total time taken for Anne to complete the 10 km journey.

Answer minutes [1]

- (c) Three minutes later, Jane started to skate behind Anne. Find the average skating speed of Jane for them to complete the journey at the same time. Leave your answer correct to 2 decimal places.

Answer km/min [2]

- 19 In the figure, AB is parallel to CD . PQ and QR bisect $\angle APR$ and $\angle CRP$ respectively.



Given that $\angle DRP = 4x^\circ$,

- (a) express angle QRP in terms of x ,

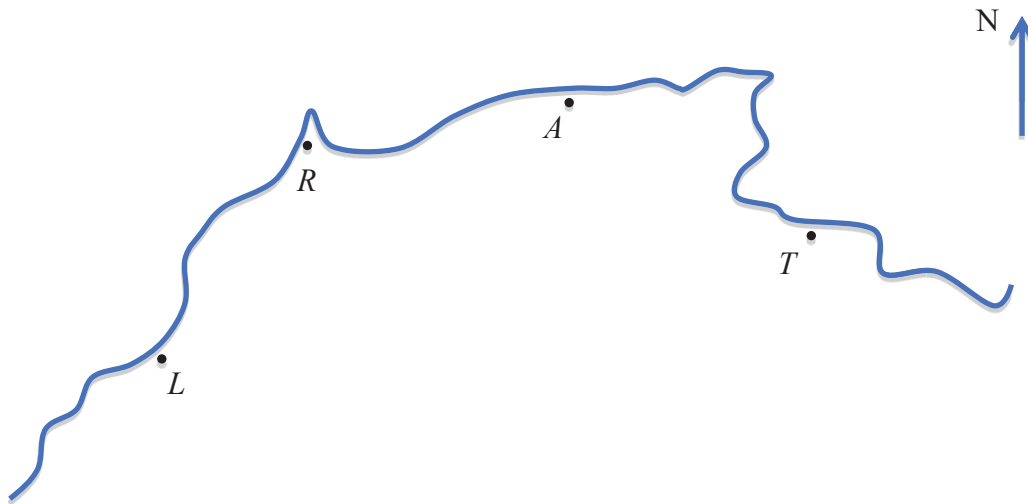
Answer [1]

- (b) show that PQ is perpendicular to QR .

Answer

[3]

- 20 The map shows four cities on the north coast of Africa. The scale of the map is such that 1 cm on the plan represents 200 km on actual ground.



- (a) Using a compass and a ruler only, construct
- (i) the perpendicular bisector of LT . [1]
 - (ii) the angle bisector of angle RAT . [1]
- (b) An ancient ruin is located at the intersection of the two bisectors in (a).
- (i) Mark and label the position of the ancient ruin with the letter **X**. [1]
 - (ii) Measure and write down the actual distance, in kilometres, of the ancient ruin from A .

Answer km [1]